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The Effect of Personal Protective Equipment on Firefighter Occupational Performance

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Abstract

Lesniak, AY, Bergstrom, HC, Clasey, JL, Stromberg, AJ, and Abel, MG. The effect of personal protective equipment on firefighter occupational performance. *J Strength Cond Res* 34(8): 2165–2172, 2020—The purpose of this study was to evaluate the effects of load carriage (LC) and LC plus respirator use (LC + self-contained breathing apparatus [SCBA]) on firefighters' work capacity to enhance our understanding of occupational demands. Twenty-one male structural firefighter recruits (age: 28.6 ± 4.3 years; height: 178.6 ± 7.2 cm; body mass: 94.1 ± 15.4 kg; body fat: 22.9 ± 6.1%) participated. Occupational performance was assessed by time to complete a simulated fire ground test (SFGT). After 2 familiarization trials, recruits performed the following SFGT conditions in a randomized order: PT (physical training clothes), LC only, and LC + SCBA. To describe within-group differences between SFGT conditions, relative difference scores were calculated as follows: % difference = ([experimental trial outcome − PT trial outcome]/PT trial outcome) × 100. Statistical differences between conditions were assessed with repeated-measures analysis of variance. The level of significance was set $p < 0.01$. Time to complete the LC + SCBA trial (345.9 ± 43.7 seconds; $p < 0.001$) and LC-only trial (331.2 ± 39.3 seconds; $p < 0.001$) were significantly greater than the PT trial (241.0 ± 33.3 seconds). Post-SFGT rating of perceived exertion was higher in the LC + SCBA trial (6.7 ± 1.7) and LC trial (6.4 ± 1.5) compared with the PT trial (4.7 ± 1.8; $p < 0.001$). Heart rate and lactate measures were similar across conditions ($p = 0.488$; $p = 0.287$). Personal protective equipment (PPE) significantly decreases the work capacity and increases the perceived effort of occupational tasks. Thus, these findings describe the additional physical demands produced by PPE and indicate that performance of firefighting tasks in an unloaded condition does not reflect work capacity in a bona fide condition.

Key Words: tactical performance, load carriage, fitness outcomes, SCBA

Introduction

Firefighting is a strenuous occupation that requires high-intensity work, resulting in prolonged periods of stress and physical exertion (34,36,37). The physical demand of performing firefighting tasks is augmented by the weight of personal protective equipment (PPE) worn (i.e., load carriage: LC) and the use of a self-contained breathing apparatus (SCBA) (12,36,37). These factors have been shown to increase metabolic demand at submaximal workloads and decrease maximal aerobic capacity in laboratory settings (2,11,34–37). However, there is limited research evaluating the effects of these factors on occupational performance.

Structural firefighters are required to perform physically strenuous occupational tasks while working in unfavorable conditions and wearing cumbersome PPE (34,36,37). The public, their colleagues, and their own lives depend on their ability to complete these tasks successfully. Firefighters' PPE consists of boots, coat, pants, hood, helmet, gloves, and an SCBA, which collectively weigh approximately 20 kg (36). Although PPE is necessary for protection, it increases the physiological burden during occupational performance because of the extra LC, increased work of breathing from the SCBA, and the microclimate created by the thermal barriers of the bunker gear (12,34,36,37). To perform occupational tasks safely and effectively, a certain level of physical fitness is required to compensate for these physiological challenges (8,23,28,32).

Load carriage and respirator use are 2 of the most prominent factors that negatively affect firefighter physical ability (12,36). The physiological impacts of LC include decreased exercise tolerance, capacity, and efficiency, and thus firefighters experience a decrease in maximal measures of workload, oxygen consumption, and power output (2,11,34–37). Load carriage also elicits greater submaximal exercise responses, including breathing discomfort, leg fatigue, rating of perceived exertion (RPE), oxygen consumption, elevated heart rate, and overall metabolic burden and physiological strain (2,11,27,34–37). In addition, LC requires increased muscular work, which is associated with an earlier onset of fatigue and decreased exercise tolerance and capacity (2,11,34–37). The overall energy expenditure of activity, as demonstrated by oxygen consumption, is increased with LC, but it is dependent on several factors including the intensity of activity, body size, location of the LC, and fitness level (26,36,37). These factors lead to an increase in energy expenditure at submaximal and maximal intensities and decreased work output. In brief, more energy is required to sustain the increased mass, allowing less energy to contribute to performing additional external work (37). Unfortunately, the majority of this research was conducted using nonfirefighter populations and was also not conducted in an occupationally relevant manner, which limits the generalizability of the results (2,11,27,34–36). Thus, there is a need to quantify the effect of LC on work capacity in a firefighter population.

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The SCBA is a necessary piece of equipment that negatively affects firefighter physical ability. The SCBA is composed of a regulator attached to an oxygen tank. The deleterious effects of the SCBA are 2-fold. First, the SCBA is carried on the firefighter's back and is secured to the torso with a harness system that uses shoulder and waist straps, similar to a backpack. The restrictive forces applied by the harness system have been shown to restrict chest wall movement and limit ventilation (12,26,27). Second, the SCBA uses a positive pressure system. Thus, the SCBA regulator requires the firefighter to produce a greater expiratory pressure to exhale. Eves et al. (12) reported that during a graded exercise test, the combination of wearing a cylinder and breathing through the SCBA decreased $\dot{V}O_2$max by 14.9% compared with performing a graded exercise test without gear and while breathing in ambient conditions. When assessed separately, the respirator decreased $\dot{V}O_2$max by 13.1%, whereas carrying the cylinder decreased $\dot{V}O_2$max by 4.8%.

Although the SCBA provides the necessary protection from smoke and other air pollutants, it has been shown to alter breathing patterns, increase the work of breathing, and significantly decrease maximal exercise performance, while increasing submaximal cardiorespiratory responses (4,12,37). This is likely to the result of a combination of the following factors: increased sensation of breathlessness, lung hyperinflation, and increased competition for available cardiac output (especially at maximal exercise) between respiratory muscles and contracting skeletal muscles, which leads to a reduction in peripheral blood flow and skeletal muscle fatigue (4,16). This resultant skeletal muscle fatigue can result in decreased exercise capacity and increased RPE (4,37). Interestingly, in clinical environments, firefighters have demonstrated greater forced expiratory volume in 1 second (FEV1), greater forced vital capacity, and increased maximum expiratory peak pressure (MEPpeak), which are perhaps respiratory adaptations to chronic SCBA use (10,31). Both thoracic LC and SCBA use, individually, have demonstrated acute decreases in maximal inspiratory and expiratory pressures, suggesting respiratory muscle fatigue is a culprit for unfavorable respiratory responses (13).

Firefighters' PPE has been shown to negatively affect physiological responses to work and exercise in laboratory settings (2,12,34,35,37). Similar to the responses observed during laboratory exercise, PPE has a negative impact on firefighter performance (36). Firefighter occupational performance is often assessed by timing the completion of occupational tasks. However, the several studies that have evaluated occupational performance did not use PPE, used parts of it, did not use a control condition, or did not recruit firefighters (23,28,32,36). Thus, there is a need to evaluate the effect of PPE on occupational performance in a controlled yet applied setting because this information would provide practitioners with specific data regarding the impact of LC and the SCBA on realistic occupational tasks. This research is critical to assess the physiological burden of PPE to allow strength and conditioning practitioners to prepare firefighters for PPE-induced work demands and determine if fire departments can accurately assess the physical work capacity of recruits without the use of PPE. Therefore, the primary purpose of this study was to quantify the effect of LC only and LC + SCBA on firefighter recruit occupational performance. It was hypothesized that both the LC + SCBA condition and LC alone would elicit a decrement in simulated fire ground test (SFGT) performance compared with a control condition (physical training [PT] clothes only) and that the combined condition would produce a greater decrement in performance than LC alone.

Methods

Experimental Approach to the Problem

The purpose of this study was to determine if LC and respirator use (SCBA) impacted SFGT performance. The SFGT is a relevant indicator of firefighter physical ability (8). This study used a randomized, repeated measures, crossover design. The primary dependent variable was SFGT time. Secondary dependent variables included blood lactate, heart rate, RPE, thermal sensation, and bodyweight lost. The independent variables were the LC and LC + SCBA conditions.

Subjects

A convenience sample of 25 metropolitan structural fire department recruits volunteered to participate in this study. The recruits' physical characteristics are displayed in Table 1. A Physical Activity Readiness Questionnaire was given to exclude recruits that have been diagnosed with cardiovascular, pulmonary, or metabolic disease or reported contraindicated signs or symptoms of these chronic diseases. All firefighter recruits provided written informed consent after a detailed explanation was provided about the aims, benefits, and risks associated with the investigation. The recruits were informed that they were free to withdraw from the study at any time, without penalty. They were told that their participation in this study did not affect their employment or probationary status. All of the procedures used in this study were approved by University of Kentucky's Institutional Review Board before the initiation of the study. Of the 25 firefighter recruits that provided informed consent, data of 21 recruits were used for analysis. Of the recruits who were not used in the data analysis, 2 subjects dropped out as a result of leaving the academy, 1 recruit dropped out because of illness, and 1 recruit dropped out because of an incurred injury that was not associated with the study procedures. The recruits had been in the

Table 1
Physical characteristics of 21 male firefighter recruits.

			95% confidence interval for mean	
	Mean ± *SD*	Range	Lower	Upper
Body mass (kg)	94.1 ± 15.4	65.5–117.3	87.1	101.1
Fat-free mass (kg)	71.7 ± 7.7	55.7–85.8	68.2	75.3
Fat mass (kg)	22.3 ± 6.1	9.8–39.2	18.2	26.5
Body mass in PPE (kg)	119.8 ± 16.2	91.3–145.9	112.4	127.2
Age (yr)	28.6 ± 4.3	21.0–36.0	26.7	30.6
Height (cm)	178.6 ± 7.2	165.4–196.0	175.3	181.9
PPE mass (kg)	25.7 ± 1.6	23.5–28.6	25	26.4
Relative body fat (%)	22.9 ± 6.1	14.4–35.8	20.1	25.7

*PPE = personal protective equipment.

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academy for 8 weeks before familiarization trials began and 11 weeks before actual data collection started.

Procedures

This study consisted of 6 testing sessions. All testing sessions took place at the fire department's training center from 1700 to 1800. Session 1 consisted of a familiarization trial on the SFGT that was performed while wearing full PPE and required the firefighter recruit to breathe through the SCBA. Session 2 consisted of a second familiarization trial of the SFGT performed with the respirator and full PPE. The order of testing sessions 3–5 was randomized and included the following SFGT conditions: PT clothes and tennis shoes only (i.e., control condition; PT condition), "LC only" (i.e., full PPE gear but not breathing through respirator), and "LC + SCBA" use (i.e., full gear and breathing through respirator). Physical training clothes were chosen as a control condition, to compare the physiological burden, perceived effort, and occupational performance of LC and LC + SCBA with an unloaded condition because fitness assessments are often conducted in PT clothes and may not accurately reflect an individual's capabilities in a bona fide–loaded condition (37). Before and after each SFGT, body mass (to account for sweat loss) and blood lactate were assessed. Immediately after completion of the SFGT, RPE and thermal sensation were recorded. Heart rate was monitored continuously with a heart rate monitor strap (Polar Electro Oy, Inc., Kempele, Finland) and transmitted to a device on the recruit's arm (ActiTrainer; Actigraph, Pensacola, FL) throughout the SFGT. Using Actilife software (ActiGraph; Version 6), the number of myocardial contractions per 15-second epochs were recorded and time stamped. These data were multiplied by 4 to express heart rate per minute. Peak heart rate was the highest heart rate obtained throughout the test, and mean heart rate was calculated throughout the trial. Any dropped data were excluded from calculations of mean heart rate. Urine specific gravity (PAL10S; Atago, Tokyo, Japan) was collected before each trial, as a measure of hydration status.

The SFGT was used as a measure of firefighter occupational physical ability. It was developed in consultation with the fire department's training officers. The test-retest reliability of the SFGT was ICC = 0.929 based on the total completion time of 3 SFGT trials (i.e., 2 familiarization trials and 1 LC + SCBA trial). Previous research has demonstrated this type of test to have similar levels of reliability (ICC = 0.937) (8). The SFGT was performed with full PPE (National Fire Protection Association, 1971; standard issued helmet, hood, coat, pants, gloves, and boots) and SCBA (Scott, Inc., Monroe, NC) totaling 25.7 ± 1.6 kg. The SFGT was composed of the following tasks, performed in order, to simulate how they are typically performed on the fire ground: stair climb, charged hose drag, equipment carry, ladder raise, forcible entry, search, and victim rescue (8). Total SFGT time and individual task times were taken using a stopwatch (Model 461; Sportline, Hazleton, PA). Recruits began by picking up and carrying one 15.24-m section of 13/4″ fire hose (mass = 22.2 kg) packaged as a high-rise hose pack up 4 flights of stairs (17 steps per flight). The recruit placed the high-rise pack on the landing and returned down the 4 flights of stairs, touching each step on the way down. Recruits were allowed to use the handrail for stability purposes only on the stair ascent. The task split time was taken when the recruit's foot touched the ground. The recruit then proceeded 15.24 m to the hose drag task. Next, the firefighter recruit performed a charged (i.e., water pressurized) hose drag task by placing the nozzle end of 1 section of 30.48 m of 1 3/4″ fire hose over their shoulder and pulled it 25 m. The split time for the hose drag was taken at the 25-m mark. The recruit then proceeded 20.7 m and performed an equipment carry task for which they carried 2 department issued 5-gallon foam buckets (mass = 20 kg each) of 62 m. The split time for the equipment carry was taken when the buckets were placed on the ground. Next, the recruit proceeded 11.2 m and performed a ladder raise task. The recruit raised a 14-ft extension ladder from the ground to a building and lowered it back to the ground using a hand-over-hand technique, touching each rung. The split time for the ladder raise was taken when the ladder was placed on the ground. The recruit then proceeded 4.4 m and completed a forcible entry task using a Keiser Force Machine Chopping Simulator (Keiser, Inc., Fresno, CA). The recruit was positioned to strike the end of a 72.7-kg steel beam with a 4-kg sledge hammer (Trusty-Cook, Indianapolis, IN). The beam was required to move 1.5 m to complete the task. Next, the recruit performed a victim search task by walking or running 7 m and climbing up a flight of 17 stairs, then performing a right hand search by crawling 35 m around the perimeter of an interior room. The search task split time was taken when the recruit returned back to the cone at the top of the stairs. Finally, the recruit performed a victim rescue task by descending back down the flight of 17 stairs and proceeded 15.6 m to a 73-kg mannequin, which was dragged 27 m to the task and SFGT finish line. The task and total SFGT was completed when the mannequin's feet crossed the finish line.

Blood lactate was measured at rest before the SFGT and 5 minutes after the test. A finger stick and universal precautions were used to obtain the lactate sample. Following the finger stick, the first drop of blood was wiped away. The second drop of blood was used for the analysis. The calibration of the blood lactate analyzer (LactatePlus; Nova Biomedical Corporation, Waltham, MA) was checked before each testing day with low (manufacturer's acceptable range: 1.0–1.6 $mmol \cdot L^{-1}$) and high (manufacturer's acceptable range: 4.0–5.4 $mmol \cdot L^{-1}$) control solutions. Rating of perceived exertion was assessed immediately following the SFGT using the 0–10 category-ratio scale for the overall feeling of exertion from the entire SFGT. This scale has been used in previous studies to investigate subjective measures of occupational physical exertion (25,27). Recruits were assessed for thermal sensation immediately following the SFGT using the Omni Thermal Sensation Scale, which has been found valid (r = 0.77) and reliable (r = 0.81) (6). It is a 1–5 category-ratio scale that uses 1 as comfortable and 5 indicating very hot.

Testing session 6 consisted of anthropometric data collection. Standing height (to the nearest 0.1 cm) was measured without shoes with a portable stadiometer (Road Rod 214; Seca, Hanover, MD). Then, body mass was measured (to the nearest 0.1 kg) without shoes with an electronic scale (TBF-521; Tanita Corporation, Arlington Heights, IL). Body composition was measured with a tetrapolar bioelectric impedance analyzer (Bodystat 1500; BIA, Ventura, CA). Specifically, electrodes were placed on the subjects' wrist, hand, ankle, and foot while lying in the supine position. Height, body mass, age, and gender were input. The recruits' fat-free mass was calculated using the following formula:

$$\begin{aligned} FFM = & -4.104 + \left(0.518 \times \text{height}^2(\text{cm})/\text{resistance}\right) \\ & + (0.231 \times \text{mass(kg)}) + (0.130 \times \text{reactance}) \\ & + 4.229(*\text{gender}), \end{aligned}$$

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where *males = 1; = Females = 0; (r = 0.986, SEE = 1.72 kg) (17).

Relative body fat percentage was then calculated (to the nearest 0.1%).

Statistical Analyses

Descriptive statistics (mean ± SD) were used to describe demographic and outcome variables. One-way repeated-measures analysis of variance were used to determine if there were differences in the absolute values for SFGT times, heart rate, blood lactate, RPE, or thermal sensation data between baseline (PT) and experimental conditions. Data were visually inspected for normality. The data were also checked for sphericity. If the sphericity assumption was violated a Greenhouse-Geiser correction was used. Least significant differences were used for follow-up pairwise comparisons. Effect sizes were calculated using Cohen's d. Pearson product moment correlations were used to assess the relationships between LC and SCBA-induced SFGT performance decrements versus RPE, thermal sensation, environmental conditions, and body mass lost. The level of significance was set at $p \leq 0.01$ to control for potential inflation of type I error because of conducting multiple statistical comparisons.

To describe within-group changes in baseline SFGT versus experimental SFGT outcomes, relative difference scores were calculated as follows: % difference = ([experimental trial outcome − baseline outcome]/baseline outcome) × 100. Delta scores were also calculated to determine the absolute difference in time between SFGT conditions as follows: experimental SFGT trial time − baseline SFGT time. Cronbach's alpha was used to assess the test-retest reliability of the SFGT. The Statistical Package for Social Sciences (SPSS, Version 22) was used for data analysis.

Results

Table 2 describes the amount of time taken to complete the SFGT in all 3 conditions and the relative difference in time to complete the total SFGT and each individual task between gear conditions. There was a significant overall effect of equipment condition on the total SFGT time ($F(2,40) = 186.05$; $p < 0.001$). Specifically, recruits took longer to complete the SFGT in the LC condition (331.2 ± 39.3 seconds; $p < 0.001$) and the SCBA condition (345.9 ± 43.7 seconds; $p < 0.001$) compared with the PT condition (241.0 ± 33.3 seconds). In addition, the time to complete SCBA condition trended toward significantly longer, when compared with the LC condition ($p = 0.020$). There were significant overall effects for condition on all SFGT task times ($p < 0.001$). Specifically, compared with the PT condition, recruits took longer to complete the SFGT during the LC and SCBA conditions on all individual tasks. In addition, recruits took longer to complete the victim rescue task ($p \leq 0.047$) in the SCBA condition compared with the LC condition. The LC condition took 38.3 ± 12.6% longer to complete the SFGT than the PT condition, whereas the SCBA condition took 44.5 ± 15.5% longer to complete the SFGT than the PT condition.

Table 3 describes the differences in physiological measures between the 3 gear conditions; RPE was significantly greater ($F(2,40) = 18.69$; $p < 0.001$) in the SCBA (6.7 ± 1.7; $p < 0.001$) and LC (6.4 ± 1.5; $p < 0.001$) conditions compared with the PT condition (4.7 ± 1.8). Thermal sensation was also significantly higher ($F(2,40) = 32.88$; $p < 0.001$) in the SCBA (3.9 ± 0.8; $p < 0.001$) and LC (3.8 ± 0.5, $p < 0.001$) conditions compared with

Table 2
Comparison of simulated fire ground test task times across equipment conditions in 21 male firefighter recruits.*

	PT			LC						SCBA					
		95% confidence interval for mean			95% confidence interval for mean		ES	Percent difference from PT			95% confidence interval for mean		ES	ES	Percent difference from PT
	Mean ± *SD*	Lower	Upper	Mean ± *SD*	Lower	Upper	LC vs. PT	Mean ± *SD*		Mean ± *SD*	Lower	Upper	CBA vs. PT	LC vs. SCBA	Mean ± *SD*
Total time (s)	241.0 ± 33.3	225.8	256.2	331.2 ± 39.3†	313.3	349.1	2.5	38.3 ± 12.6		345.9 ± 43.7†	326	365.8	2.7	0.4	44.5 ± 15.5
Stair climb (s)	42.0 ± 7.6	38.6	45.5	66.7 ± 10.2†	62.1	71.3	2.7	60.7 ± 21.2		70.1 ± 12.1†	64.6	75.6	2.8	0.3	69.2 ± 26.9
Hose drag (s)	21.5 ± 3.7	19.8	23.2	32.2 ± 4.6†	30.0	34.3	2.5	52.1 ± 24.4		33.5 ± 5.3†	31.1	35.9	2.6	0.3	58.3 ± 26.4
Equipment carry (s)	42.5 ± 5.6	39.9	45	59.0 ± 5.8†	56.4	61.6	2.9	40.3 ± 16.2		60.6 ± 5.8†	57.9	63.2	3.2	0.3	44.3 ± 17.8
Ladder raise (s)	15.7 ± 2.4	14.6	16.8	20.0 ± 2.7†	18.7	21.2	1.7	28.0 ± 12.9		20.8 ± 2.9†	19.5	22.1	1.9	0.3	33.2 ± 14.5
Forcible entry (s)	41.4 ± 9.2	37.2	45.6	49.1 ± 9.5†	44.8	53.5	0.8	21.9 ± 25.3		50.7 ± 11.1†	45.6	55.8	0.9	0.2	25.6 ± 27.4
Search (s)	43.3 ± 9.2	39.1	47.5	54.8 ± 8.2†	51.1	58.6	1.3	29.3 ± 18.9		57.6 ± 9.0†	53.5	61.6	1.6	0.3	36.1 ± 22.2
Victim drag (s)	34.7 ± 6.4	31.8	37.6	49.4 ± 7.7†‡	45.9	52.9	2.1	44.5 ± 15.5		52.7 ± 7.7†	49.5	56.2	2.6	0.4	54.5 ± 23.1
Heat index (°C)	23.1 ± 4.2	21.3	24.9	25.9 ± 2.4	24.9	26.9	0.8			24.9 ± 5.1	22.7	27.1	0.39	0.3	

*PT = wearing physical training clothes; LC = wearing full gear but not breathing through SCBA; SCBA = wearing turnout gear and breathing through respirator; ES = effect size (Cohen's d); SCBA = self-contained breathing apparatus; LC = load carriage; PT = physical training.
†Significant difference compared to the PT condition ($p < 0.01$).
‡Significant difference between LC vs. SCBA conditions ($p < 0.01$).

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Table 3
Physiological and psychological measures taken in 3 conditions of the simulated fire ground test in 21 male firefighter recruits.*

	PT			LC				SCBA					
		95% confidence interval for mean			95% confidence interval for mean		ES		95% confidence interval for mean		ES	ES
	Mean ± *SD*	Lower	Upper	Mean ± *SD*	Lower	Upper	LC vs. PT	Mean ± *SD*	Lower	Upper	SCBA vs. PT	LC vs. SCBA
Average heart rate (b·min^{-1})	161.3 ± 12.7	155.9	166.7	161.6 ± 11.2	156.8	166.4	0.0	162.8 ± 13.4	157.1	168.5	0.1	0.1
Average heart rate (%)	84.3 ± 5.7	81.7	86.9	84.3 ± 5.0	82	86.6	0.0	85.0 ± 6.5	82	87.9	0.1	0.1
Peak heart rate (b·min^{-1})	171.8 ± 13.9	165.9	177.7	173.6 ± 12.4	168.3	178.9	0.1	174.7 ± 13.4	169	180.4	0.2	0.1
Peak heart rate (%)	89.8 ± 6.1	87	92.6	90.6 ± 5.7	88	93.2	0.1	91.1 ± 6.2	88.3	93.9	0.2	0.1
RPE	4.7 ± 1.8	3.8	5.4	6.4 ± 1.5†	5.7	6.9	1.0	6.7 ± 1.7†	5.9	7.5	1.1	0.2
Thermal sensation	2.2 ± 0.9	1.8	2.6	3.8 ± 0.5†	3.6	4.2	2.2	3.9 ± 0.8†	3.5	4.2	1.9	0.1
BM lost (kg)	0.1 ± 0.2	0.03	0.21	0.3 ± 0.28	0.1	0.4	0.6	0.3 ± 0.23	0.2	0.4	0.8	0.1
Resting lactate (mmol·L^{-1})	1.4 ± 1.2	0.9	1.9	1.4 ± 0.6	1.2	1.7	0.1	1.6 ± 0.7	1.3	1.9	0.2	0.2
Post lactate (mmol·L^{-1})	12.4 ± 4.2	10.5	14.3	11.1 ± 2.4	10.1	12.2	0.4	11.1 ± 2.9	9.8	12.4	0.4	0

*PT = wearing physical training clothes; LC = wearing full gear but not breathing through SCBA; SCBA = wearing turnout gear and breathing through respirator; RPE = rating of perceived exertion; BM lost = body mass lost; average heart rate % = percent age predicted maximum; ES = effect size (Cohen's *d*), SCBA = self-contained breathing apparatus; LC = load carriage; PT = physical training.
†Significant difference compared to the PT condition ($p < 0.01$).

the PT condition (2.2 ± 0.9). The amount of body mass lost from the SFGT was not significantly different between the conditions ($F(2,30) = 3.51$; $p = 0.039$). Despite differences in thermal sensation responses between PT vs. LC and SCBA conditions, the decrement in the absolute SFGT time did not correlate with the change in thermal sensation for LC ($r = 0.075$; $p = 0.745$) or for SCBA ($r = -0.078$; $p = 0.736$). Likewise, the differences in body mass lost and the decrement in the absolute SFGT did not correlate for LC ($r = -0.243$; $p = 0.289$) or for SCBA ($r = 0.123$; $p = 0.596$). The decrement in the absolute SFGT time did not correlate with the change in RPE for LC ($r = -0.168$; $p = 0.467$) or for SCBA ($r = 0.148$; $p = 0.523$). There was no difference in urine-specific gravity before the SFGT between all trials ($F(1.372, 24.702) = 0.711$; $p = 0.450$). Specifically, urine-specific gravity was 1.0209 ± 0.00826 in the PT condition, 1.0178 ± 0.00851 in the PPE condition, and 1.0181 ± 0.01042 in the PPE + SCBA. Serious dehydration is considered greater than 1.030 (7).

Discussion

The purpose of this study was to determine if there was an effect of LC and LC plus SCBA use on occupational performance in firefighter recruits. It took recruits significantly longer to complete the SFGT in the LC and SCBA conditions compared with the PT condition, indicating that the mass of the gear and positive pressure respirator inhibited work efficiency. Furthermore, the victim rescue task took longer to complete in the SCBA condition compared with the LC condition, indicating that the respirator produced an additional deleterious effect on work efficiency.

The LC condition took 38.3 ± 12.6% longer than the PT condition, whereas the SCBA condition took 44.5 ± 15.5% longer. This is similar to the 27% decrement in firefighter physical ability produced by PPE reported by Taylor et al. (36). Similarly, using a military population loaded with 42 kg of gear, Mala et al. (22) found a 31% decrease in performance on an anaerobic obstacle course that included a sprint and a victim drag task. The current study elicited an average decrement of 1.5 ± 0.43% per kg of PPE, which is a smaller decrement than has been previously reported in the literature, indicating that body armor results in an average of 2.4–3.5% decrement per kg of body armor in military populations in agility and obstacle course proficiency, with a lesser decrement (1.5% per kg) reported in treadmill walking or rucking performance (9,18–20). Military tasks are broken down into material handling and movement tasks. Material handling tasks involve more fine motor movements (i.e., rifle firing and loading). A 30% decrement in military task performance has been demonstrated with only a 4.5-kg protective vest and is attributed to the mass of the vest but also the functional obstruction of the stiffness of the vest (9,14,18). The smaller relative decrement seen in this study may be the result of the SFGT incorporating more gross motor tasks and fewer fine motor skill tasks than the previously mentioned studies. In both PPE conditions, the stair climb task was affected to the greatest extent because a 60.7 ± 21.2% decrement in performance in the LC condition and 69.2 ± 26.9% decrement in the SCBA condition were noted. Thomas (38) reported much lower decrements than the current study in stair ascent/descent (14%) and victim drag (15.6%) tasks with SWAT operators. The increased decrements reported in the current study are likely the result of the greater number of stairs ascended and descended (4 flights vs. 1 flight), the requirement to drag a mannequin a greater distance (27 vs. 23 m), and the greater LC requirement overall (25.7 vs. 14.2 kg) (38). In the present study, the search task took 29.3 ± 18.9% longer in the LC condition,

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similar to Thomas (38) who reported a 34.3% decrement. These findings support the contention that the performance decrement induced by LC is affected by the type of task performed, the magnitude of load carried, and the placement and material characteristics (i.e., stiffness) of the load (14,37).

The results from the current study suggest that together, the respirator and LC have a compounded effect on simulated work capacity. This is similar to what is reported in the literature during laboratory tests. Eves et al. (12) demonstrated that carrying and breathing through the SCBA decreased peak power output on a maximal treadmill graded exercise test by 11.6%, whereas carrying the cylinder itself decreased peak power by 6.4% and breathing through the regulator without the LC decreased peak power by 4.8%. The subjects in the aforementioned study were not wearing full PPE, instead wearing only shorts, running shoes, a t-shirt, and National Fire Protection Association standard jacket, as their variable of interest was the impact of the SCBA components. This may explain the difference in findings when comparing the decrement of the SCBA condition with this study. However, it is challenging to compare the present study's findings across the literature as the authors are not aware of any literature comparing the effects of the SCBA + LC with a LC condition while performing occupational tasks in a sample of recruits.

The physiological measures taken during and following the SFGT were relatively similar between the 3 conditions despite taking longer to complete the SFGT in the PPE conditions. Specifically, recruits' mean and peak heart rate values were similar between conditions in absolute terms (Table 3). When calculated as a percentage of HRmax, peak heart rate was slightly higher in the SCBA condition. In addition, the mean heart rates from the current study in the SCBA condition (162.8 ± 13.4 $b \cdot min^{-1}$; 85.0 ± 6.5% HRmax) were similar to those from the study by Dennison et al. (8) who used a similar SFGT and reported an average heart rate of 165.5 ± 12.5 $b \cdot min^{-1}$ (87.6 ± 5.9% HRmax). Furthermore, despite similar heart rate values between conditions, RPE and thermal sensation were significantly higher in the SCBA and LC conditions. This finding is supported in the literature, where research indicates that the same task is perceived as being more difficult to complete while wearing gear, because of an increased metabolic demand and increases in skin and microclimate temperatures (3,11,36).

Thermal regulation is yet another component of PPE that leads to increased cardiovascular strain during firefighter tasks. Heat transfer is gradient dependent, and the hot microclimate created by the PPE does not allow heat to dissipate effectively (34,37). The higher thermal scores noted in the SCBA and LC conditions demonstrate the reduced ability of the gear to dissipate heat, leading to the development of a hot microclimate (3). The impaired ability to thermoregulate because of the PPE has been shown to decrease exercise tolerance, capacity, and power output as and increase the cardiovascular strain at submaximal conditions, which is likely the result of the competition for blood flow among the working muscles and the shunting of blood to the skin in an attempt to reduce core body temperature (1,3,33,34,37). Although not statistically significant, the negative effects of the hot microclimate within the gear were seen in the loss of body mass in the SCBA condition ($p = 0.027$), with a similar result in the LC condition ($p = 0.036$; Table 3). The recruits lost an average of 0.284 kg (range, 0–0.82 kg) in the SCBA condition, which ranged from 0 to 0.99% of their body mass. Similarly, Caldwell et al. (5) reported that subjects lost 1.65–2.19% of their body mass following 2.5 hours of low-intensity activity in a hot humid environment while wearing body armor. The amount of body mass lost because of perspiration in this study is concerning given that the occupational tasks only took 5–6 minutes to complete. It is suggested that body mass losses of >2% impacts physical and mental performance (30). Dehydration leads to decreased fluid volume, which can increase cardiovascular strain because of decreased venous return and increased work of the heart to maintain cardiac output and may result in the depletion of electrolytes and increase the risk for heat illnesses (30,34,37).

The post-SFGT blood lactate concentrations were relatively high and similar across PPE conditions (i.e., 11.1–12.4 $mmol \cdot L^{-1}$), reflecting the anaerobic demands of performing multiple fire ground tasks, regardless of PPE utilization. Despite similarities in blood lactate concentrations across PPE conditions, RPE was significantly higher and work capacity was reduced when PPE was used. There are reports in the literature of other cases where LC (i.e., body armor) has increased RPE and decreased work capacity yet elicited no difference in blood lactate across conditions (29). Thus, in the present study, these results are likely the result of the fact that in non-PPE conditions, individuals can perform occupational tasks more quickly, are not exposed to the ventilatory burden of the respirator, or the thermoregulatory strain produced by turnout gear and therefore perceive the effort to be less, despite still relying heavily on anaerobic metabolism. The magnitude of the post–blood lactate values noted in the present study (i.e., 11.1–12.4 $mmol \cdot L^{-1}$) are similar to those reported in other studies using simulated fire ground tasks. Von Heimburg et al. (39) found blood lactate concentrations of 13 ± 3 $mmol \cdot L^{-1}$ after tasks involving stair climbing and a victim drag, and Dennison et al. (8) found values of 11.8 ± 3.2 $mmol \cdot L^{-1}$ after performing a similar SFGT wearing PPE. Collectively, these findings support the contention that these high-intensity tactical activities are rather anaerobic in nature (15,21,22,38,39). Harvey et al. (15) further supported this by demonstrating that by the end of the first task in a SFGT, $\dot{V}CO_2$ had already exceeded $\dot{V}O_2$, likely because of buffering of hydrogen ions in response to the sustained muscular contractions during such activities. The literature is mixed as to the importance of anaerobic power vs. capacity because it may be argued that in emergency firefighting operations that require swift action, anaerobic capacity may be of less importance than maximal anaerobic power (15,39). In addition, it has been suggested that having a larger maximal aerobic capacity could allow a reduced contribution of anaerobic metabolism (32) (Figure 1).

There are several limitations to the present study including subject variability and differences in environmental conditions. The LC condition was designed to remove the effects of the positive pressure of the respirator but maintain the effects of LC. Therefore, the mask, helmet, and tank were worn, but the regulator was disconnected from the face mask in the LC condition. The hood was also removed in the LC condition to prevent the face shield from fogging and obstructing visibility. Some of the recruits stated that the LC condition felt more difficult than the SCBA condition. Because the face shield was still partially covering their face, it may have resulted in a feeling of dyspnea and slowed down air turnover. In addition, thoracic loading alone has been shown to decrease exercise tolerance by restricting chest wall movement and therefore ventilation (24,26,27). Although the LC condition was aimed at evaluating the effect of the LC and microclimate of the PPE, the SCBA cylinders were still full of air and strapped to the subjects' back. Tidal volume has been shown to decrease in both traditional LC and chest strapping, which simulates the chest constriction without the actual load (24). In addition, the

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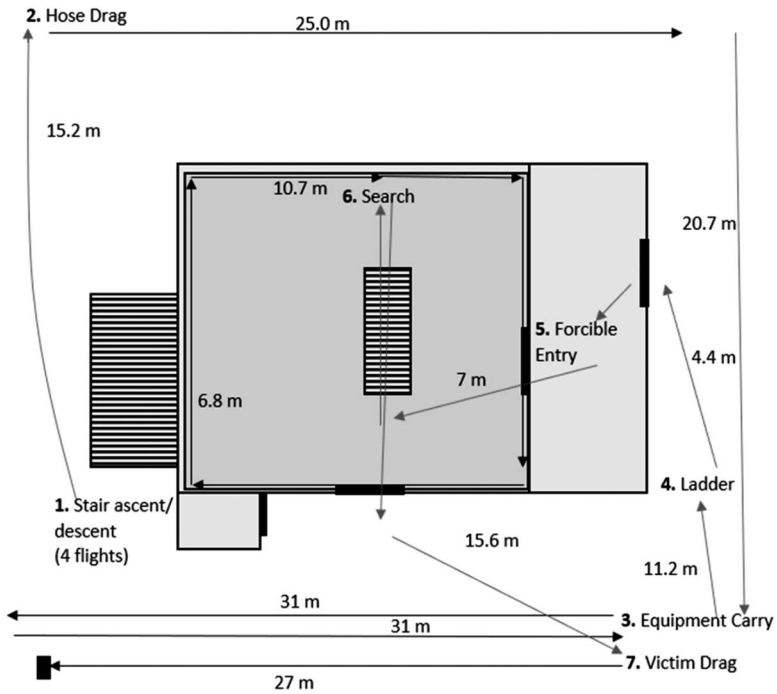

Figure 1. Schematic of simulated fire ground test dimensions.

SCBA trial could not isolate the decrement caused specifically by the respirator alone. As the current study used fire academy recruits as opposed to incumbent firefighters, it is pertinent to justify their inclusion and compare their physical characteristics and physical fitness levels to incumbent firefighters previously described in the literature. First, the recruits in the current study had been in the academy wearing PPE and performing fire ground tasks for 8 weeks before familiarization trials began and 11 weeks before actual data collection started; thus, they were accustomed to wearing PPE. Regarding demographic and physical characteristics, the recruits were younger (28.6 ± 4.3 years) compared with the incumbent firefighters reported in investigations conducted by Michaelides et al. (23) (33 ± 7 years) and Rhea et al. (28) (35 ± 6 years). On average, the recruits in the present study had a lower body fat percentage (22.9 ± 6.1%) compared with the incumbents who participated in a study by Michaelides et al. (23) (23 ± 6%) but higher than incumbents in a study by Rhea et al. (28) (17 ± 4%) (23,28). The current recruits also had similar handgrip strength (right hand: 62.3 ± 7.2 kg; left hand: 59.5 ± 5.5 kg) compared with incumbents reported by Rhea et al. (28) (right hand: 60.9 ± 8.6 kg; left hand: 57.8 ± 8.4 kg). Furthermore, Dennison et al. (8) completed a similar SFGT using incumbent firefighters and reported an overall SFGT time and task times very similar to those reported in the current study. Thus, the current group of firefighter recruits had some but not extensive experience with PPE. In addition, they tend to be younger than some incumbent firefighters but have similar body composition and upper-body strength levels as incumbent firefighters reported in the literature.

In conclusion, this study quantified the decrement in firefighter occupational physical ability caused by LC and LC + SCBA in firefighter recruits. Although necessary, firefighting PPE produces tremendous performance decrements as the LC condition elicited a 38% decrement in performance, the respirator produced an additional 4.6% decrement, and the total PPE elicited a 44.5% decrement. Finally, this study demonstrated that performance in PT attire is not an accurate depiction of occupational physical ability while wearing PPE.

Practical Applications

This study quantified the negative impact on performance and increased physiological stress caused by both LC and the SCBA while completing occupational tasks in a firefighter recruit population. Practitioners should use this knowledge to create training programs that elicit the intensity of the physiological stress noted on the fire ground while wearing PPE. Furthermore, training programs should incorporate PPE into select aspects of a comprehensive periodized program to enhance the occupational specificity of training (i.e., perform simulated fire ground tasks in PPE). In addition, practitioners and fire service personnel should be aware that because not all recruits were equally impacted by the LC or LC + SCBA, occupational physical ability in an unloaded condition may not accurately reflect occupational performance in gear. Therefore, it is pertinent to assess occupational physical ability while wearing the full PPE ensemble.

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